

Human Capital Theory: Applications to Education and Training

Chapter 9

Prepared by Dr. Mir Ahasan Kabir

York University

Learning Objectives

- Understand how investments in education and training can be analyzed as an **economic investment decision**.
- Recognize that the **opportunity cost of time** is often the largest cost of education.
- Understand why comparing wages of educated and less educated workers can be misleading due to:
 - ability bias
 - labour market signalling
- Explain why the **return to education** is large and increasing in Canada.
- Discuss when workers or employers should finance training programs.

Explanation:

Human capital theory treats education similarly to investments in physical capital. Individuals compare the **costs of schooling** with the **future benefits in higher earnings**.

Main Questions

- Why are more educated workers paid more than less educated workers?
- What determines the market rate of return to education?
- If education is so profitable, why doesn't everyone pursue a PhD?
- Do wages reflect the productivity gained through education or simply signal ability?
- What roles do signalling and screening play in labour markets?

Explanation

These questions motivate the theory of human capital. Economists attempt to determine whether education actually **raises productivity** or simply acts as a **signal of ability**.

Human Capital Theory

Definition

Human capital refers to the knowledge, skills, and abilities acquired through education, training, and experience that increase productivity and earnings.

Key ideas:

- Investments in education involve **costs today**.
- These investments generate **higher earnings in the future**.
- Rational individuals invest if the **present value of benefits exceeds costs**.

Example

A student who attends university for four years sacrifices income during those years but expects higher wages over the rest of their career.

Costs and Benefits of Human Capital Investment

Important components include:

- **Opportunity cost:** forgone earnings while studying.
- **Private costs and benefits:** costs and gains experienced by the individual.
- **Social costs and benefits:** effects on society as a whole.
- **Real costs:** resources used for education (teachers, buildings).
- **Transfer costs:** financial transfers such as taxes or subsidies.

Example

Tuition is a private cost to students but may be subsidized by the government, shifting some costs to taxpayers.

Explanation

The diagram compares two lifetime income streams:

- income if the individual starts working immediately
- income if the individual attends school first

Education typically reduces income early in life but increases income later.

Age Earnings Profiles

Empirical evidence shows that:

- Earnings increase with age and experience.
- The increase occurs at a decreasing rate.
- Workers with more education earn consistently higher wages.

Example

University graduates typically start earning later but surpass high school graduates within several years.

Optimal Human Capital Investment

Assumptions used in the model:

- Education has no direct consumption value.
- Work hours are fixed.
- Income streams are known with certainty.
- Perfect capital markets allow borrowing and lending.

Economic Idea

Individuals choose the level of schooling that maximizes the **present value of lifetime earnings**.

Net Present Value of Education

For an 18-year-old deciding whether to attend college:

$$PV = \sum_{t=18}^T \frac{Y}{(1+r)^t}$$

where:

- Y = annual income
- r = interest rate
- T = retirement age

Education is worthwhile if the **present value of higher future earnings exceeds the cost of schooling**.

Marginal Costs and Benefits

Cost of one more year of schooling:

$$MC = Y + D$$

where

- Y = forgone earnings
- D = direct costs (tuition, books)

Benefit:

$$MB = \Delta Y$$

which represents the increase in future income.

Optimal Education Rule

Education should increase until:

$$MB = MC$$

or equivalently

$$i = r$$

where

- i = internal rate of return to education
- r = market interest rate

Interpretation

If the return to education exceeds the interest rate, additional schooling is profitable.

Implications of the Theory

- Human capital investment occurs mainly early in life.
- Individuals expecting shorter careers invest less.
- Progressive income taxation may reduce incentives to invest in education.

Example

Workers nearing retirement rarely invest in long-term education programs.

Factors Influencing Education

Key determinants:

- **Income taxes:** progressive taxes reduce returns to education.
- **Student loans:** reduce borrowing constraints and increase schooling.
- **Government subsidies:** encourage investment in human capital.

Worked Example: Returns to Education

Mary can earn \$30,000 immediately.

Interest rate = 5%.

If education raises income to \$33,000:

$$i = \frac{\Delta Y}{Y + D}$$

If direct costs are zero:

$$i = \frac{3000}{30000} = 10\%$$

Education is profitable because the return exceeds the interest rate.

Signalling Theory

Education may act as a signal of worker productivity rather than increasing productivity itself.

Employers use education to screen workers because:

- ability is difficult to observe directly
- educational achievement may reveal ability

Empirical Evidence: Education and Earnings

Key patterns:

- Earnings rise with experience.
- Workers with more education earn significantly more.
- The earnings gap widens with age.

Human Capital Earnings Function

Economists estimate returns to education using:

$$\ln Y = \alpha + rS + \beta_1 EXP + \beta_2 EXP^2 + \epsilon$$

Where:

- Y = earnings
- S = years of schooling
- EXP = experience

The coefficient r estimates the **return to education**.

Ability Bias

Estimating returns to education is complicated because:

- ability
- motivation
- perseverance

may affect both education and wages.

Researchers address this using:

- twin studies
- natural experiments
- compulsory schooling laws

Returns to Education and Inequality

Evidence suggests:

- returns to education have increased since the mid-1990s
- higher returns contribute to rising wage inequality
- technological change increases demand for skilled workers

Training

Two types of training:

General training

- transferable across firms
- workers usually pay the cost

Specific training

- valuable only within one firm
- firms often pay for the training

Government Training Programs

Government intervention may be justified when:

- private markets underinvest in training
- information is imperfect
- disadvantaged workers face borrowing constraints

Training subsidies may increase employment and earnings.

Chapter Summary

- Education is an investment in human capital.
- Individuals choose schooling by comparing costs and benefits.
- Education may increase productivity or signal ability.
- Empirical evidence shows strong returns to education.
- Training programs and government policies influence human capital investment.

Thank You